

Wieferich–Ulm Signatures and the Classification of Compact Arithmetic Packet Bases

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Abstract

For a rational prime p , let $U_p = \prod_{\ell \neq p} \mathbf{Z}_\ell^\times$, let $e_p : \widehat{\mathbf{Z}} \rightarrow U_p$ be the continuous exponent map $a \mapsto p^a$, and put $B_p = U_p / e_p(\widehat{\mathbf{Z}})$. These compact abelian groups arise as arithmetic packet bases, but their topological isomorphism types are not visible from the omitted coordinate alone. This paper computes the complete primary structure of the unmarked groups B_p . For every rational prime r , define

$$\kappa_r(p) = \begin{cases} 0, & r = p, \\ v_r(p^{r-1} - 1) - 1, & r \neq p \text{ and } r \text{ is odd}, \\ v_2(p^2 - 1) - 3, & r = 2 \text{ and } p \text{ is odd}. \end{cases}$$

If $B_{p,(r)}$ is the pro- r Sylow factor and $A_{p,r} = \overline{\text{Tor}(B_{p,(r)})}$, then $A_{p,r} \cong \prod_{n \geq 1} (C_{r^n})^{\aleph_0}$ and $B_{p,(r)} / A_{p,r} \cong C_{r^{\kappa_r(p)}}$. The proof combines an exceptional-primary triangular kernel split with an off-local exact-order Kummer–Chebotarev construction. The latter yields roots inside the relevant restriction kernel, not merely in an ambient Pruefer group, and determines the full finite and transfinite Ulm sequence. Consequently,

$$B_p \cong_{\text{top}} B_q \iff \kappa_r(p) = \kappa_r(q) \text{ for every prime } r.$$

At $r = 11$ this separates B_2 from B_3 . The theorem concerns the bare compact quotient only: it does not recover marked coordinates, actual packet topology, measure, flow, trace, operator, or determinant data. Injectivity of the full signature $p \mapsto (\kappa_r(p))_r$ is not resolved here.

Keywords: profinite abelian groups; Ulm invariants; Wieferich defects; Pontryagin duality; compact arithmetic packet bases

中文摘要

对有理素数 p , 令 $U_p = \prod_{\ell \neq p} \mathbf{Z}_\ell^\times$, 以连续指数映射 $e_p : \widehat{\mathbf{Z}} \rightarrow U_p$ 定义裸紧商群 $B_p = U_p / e_p(\widehat{\mathbf{Z}})$ 。这些群来自算术 packet 基底, 但其裸拓扑同构型不能由被删去的坐标直接读出。本文先证明 e_p 为闭嵌入并建立规范主分解, 再计算 B_p 的全部主分量与 Ulm 不变量。对每个素数 r , 局部缺陷 $\kappa_r(p)$ 在 $r = p$ 时为零, 在奇素数 $r \neq p$ 时为 $v_r(p^{r-1} - 1) - 1$, 在 $r = 2$ 、 p 为奇数时为 $v_2(p^2 - 1) - 3$ 。若 $A_{p,r}$ 是 B_p 的 pro- r 分量中扭元子群的闭包, 则 $A_{p,r} \cong \prod_{n \geq 1} (C_{r^n})^{\aleph_0}$, 且特征商群为 $C_{r^{\kappa_r(p)}}$ 。对角分支不能从原始因子直接

推出有界角色满射，故以齐次三角分解和 Kulikov 吸收处理；非对角分支则在有限 Galois 复合扩张中对 Frobenius 共轭类应用 Kummer–Chebotarev 构造，同时控制素数坐标和 p 的精确阶，并以有限坐标修正把各深度的根留在限制核内。这给出完整有限及超限 Ulm 序列。最终， B_p 与 B_q 拓扑同构，当且仅当所有 κ_r 坐标相等； $r = 11$ 已区分 B_2 与 B_3 。结论仅属于无标记裸紧群，不转移到实际 packet、测度、流、迹、算子或行列式。证明还严格区分裸群不变量与标记、测度及动力学对象的所有权，防止将分类结论越界转移；该签名对全部素数是否单射，本文不作结论。

中文关键词：前有限阿贝尔群；Ulm 不变量；Wieferich 缺陷；Pontryagin 对偶；算术 packet 基底

1 Introduction and main results

The compact groups studied here originate in Deninger’s arithmetic-dynamical packet construction. In the notation used below, the source supplies the quotient

$$B_p = \widehat{\mathbf{Z}}_{(p)}^\times / p^{\mathbf{Z}} \cong \text{Aut}(\overline{\mathbf{F}}_p^\times) / \text{Aut}(\overline{\mathbf{F}}_p)$$

and a choice-dependent packet fibration over it [3, equations (38)–(40), Section 6, and Theorem 6.1]. That provenance does not identify the topology of an actual packet quotient with the compact topology of B_p , nor does it determine the Sylow or Ulm structure. The present question is therefore intrinsic and purely topological: what information about p is retained by the unmarked compact abelian group B_p ?

The answer is a full classification by local Wieferich depths. The depth at the missing primary coordinate is zero. At an odd off-local prime r , it is the excess r -adic valuation of $p^{r-1} - 1$ beyond Fermat’s forced factor. At $r = 2$, the sign and principal-unit factors must be separated, producing $v_2(p^2 - 1) - 3$. These numbers are assembled into the signature $\kappa(p) = (\kappa_r(p))_r$.

Theorem 1.1 (Primary torsion-closure structure). *For every pair of rational primes p, r , let $B_{p,(r)}$ be the pro- r Sylow factor of B_p and put*

$$A_{p,r} = \overline{\text{Tor}(B_{p,(r)})}.$$

Then

$$A_{p,r} \cong_{\text{top}} P_r, \quad B_{p,(r)} / A_{p,r} \cong_{\text{top}} C_{r^{\kappa_r(p)}}, \quad P_r = \prod_{n \geq 1} (C_{r^n})^{\aleph_0}. \quad (1)$$

In particular, $\kappa_r(p)$ is intrinsic to the unmarked compact group.

Theorem 1.2 (Complete classification). *For rational primes p and q ,*

$$B_p \cong_{\text{top}} B_q \iff \kappa_r(p) = \kappa_r(q) \text{ for every rational prime } r. \quad (2)$$

Corollary 1.3 (An explicit separation). *B_2 and B_3 are not topologically isomorphic.*

Three transitions are load-bearing. First, the diagonal branch $r = p$ does not satisfy the bounded-character surjectivity that one might try to infer from primitive divisors. It requires a separate homogeneous triangular split and a Kulikov subgroup argument. Second, for $r \neq p$ the proof forces both $v_r(\ell - 1) = m$ and $v_r(\text{ord}_\ell(p)) = m$. Divisibility of the order is not enough. Third, ambient Pruefer divisibility does not by itself compute the infinite-height subgroup of the kernel; a saturated finite-coordinate correction is needed to construct roots inside that kernel.

The local calculations also clarify the relation to the larger flow-systems program. Route A evaluates arithmetic dynamical systems with an intrinsic clock, periodic-orbit structure, and a specified dynamical zeta or Fredholm determinant. Route B requires a Hilbert space, an operator with domain, an exact arithmetic trace, and a completed-zeta determinant belonging to one construction. This paper supplies none of those objects. The Route-A required-input screen is therefore NOT_TESTABLE; no A0–A4 tuple or overall Route-A verdict is assigned. Route-B advancement is not authorized, and its exact overall status for this object is ROUTE_B_NOT_TESTABLE.

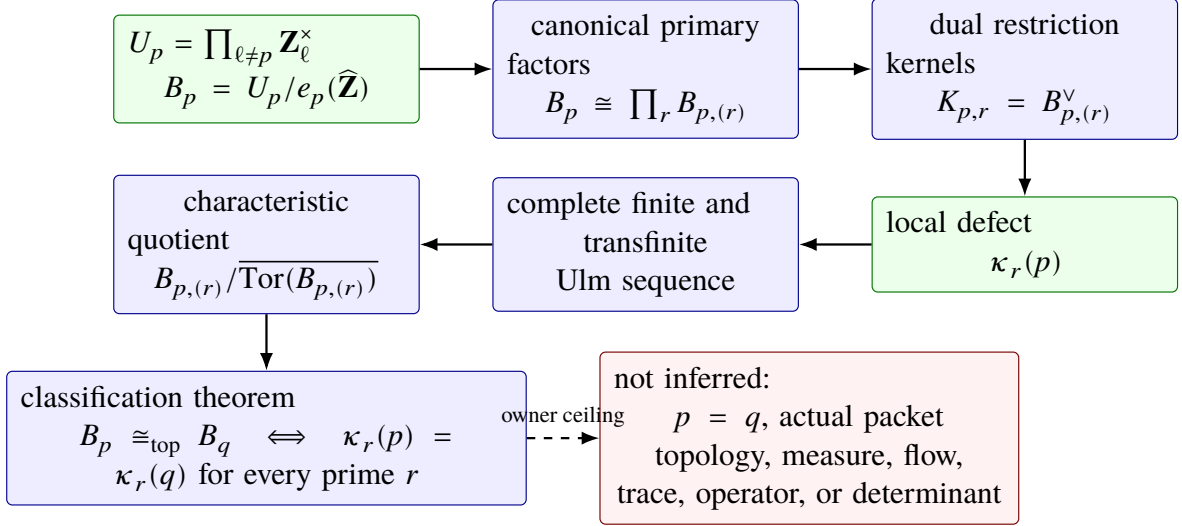


Figure 1: Proof architecture and owner ceiling. Arithmetic exact-order data enter through the restriction kernels, the complete Ulm ledger determines the primary factor, and the unrestricted product gives the global classification. The dashed arrow marks conclusions that are not licensed by the theorem.

The paper proceeds from the compact owner and duality conventions to the closed exponent embedding, the local restriction maps, the diagonal and off-diagonal kernels, the full Ulm sequence, and the compact classification. The final sections separate marked from unmarked information and state the route, computational, and publication limits.

2 Compact packet bases and Ulm conventions

All compact groups are Hausdorff and all groups are abelian. Discrete primary groups are written additively. For a prime r , write C_{r^n} for the cyclic group of order r^n , set $C_{r^0} = 1$, and write C_{r^∞} for the Prüfer r -group. Pontryagin duals are denoted by $G^\vee$.

For a rational prime p , define

$$U_p = \prod_{\ell \neq p} \mathbf{Z}_\ell^\times, \quad e_p : \widehat{\mathbf{Z}} \rightarrow U_p, \quad a \mapsto p^a, \quad (3)$$

and put

$$H_p = e_p(\widehat{\mathbf{Z}}), \quad B_p = U_p / H_p. \quad (4)$$

The meaning of p^a is coordinatewise: the ordinary map $n \mapsto p^n$ from $\mathbf{Z}$ to each $\mathbf{Z}_\ell^\times$ extends continuously to $\widehat{\mathbf{Z}}$. The continuity and injectivity of the product map are proved in [section 3](#).

Lemma 2.1 (Exact dual sequence). *If H is a closed subgroup of an abelian profinite group G , restriction gives an exact sequence of discrete groups*

$$0 \rightarrow (G/H)^\vee \rightarrow G^\vee \rightarrow H^\vee \rightarrow 0. \quad (5)$$

For a closed subgroup $A \leq G$,

$$A^\vee \cong G^\vee / A^\perp, \quad (G/A)^\vee \cong A^\perp. \quad (6)$$

Proof. A character of G/H pulls back precisely to a character of G that annihilates H . For surjectivity of restriction, let $\chi : H \rightarrow \mathbf{R}/\mathbf{Z}$ be continuous. Its image is finite in the profinite setting, and its kernel

is open. Choose an open subgroup $V \leq G$ with $V \cap H \subseteq \ker \chi$. The character then factors through the subgroup $H/(V \cap H)$ of the finite abelian group G/V . Characters of subgroups of finite abelian groups extend to the ambient finite group, for example by cyclic primary decomposition and divisibility of $\mathbf{R}/\mathbf{Z}$. Pulling an extension back to G proves surjectivity. The annihilator identities follow from the same correspondence. $\square$

Lemma 2.2 (Canonical primary factors). *Every abelian profinite group has a canonical unrestricted product decomposition*

$$G \cong_{\text{top}} \prod_r G_{(r)}, \quad (7)$$

where $G_{(r)}$ is the unique pro- r Sylow factor. Each factor is characteristic, and closed subgroups and quotients respect the decomposition.

Proof. Every finite abelian quotient is canonically the product of its primary Sylow subgroups, compatibly with transition maps. Taking the inverse limit gives (7). Uniqueness implies characteristicity. The primary idempotents in finite quotients restrict to a closed subgroup and descend through a compact quotient map. $\square$

Define

$$P_r = \prod_{n \geq 1} (C_{r^n})^{\aleph_0}, \quad S_r = P_r^\vee = \bigoplus_{n \geq 1} (C_{r^n})^{(\aleph_0)}. \quad (8)$$

For a discrete r -group K , use the transfinite filtration

$$r^0 K = K, \quad r^{\alpha+1} K = r(r^\alpha K), \quad r^\lambda K = \bigcap_{\alpha < \lambda} r^\alpha K \quad (9)$$

at a limit ordinal λ . In particular, $r^\omega K = \bigcap_{m \geq 0} r^m K$. With $J[r] = \{x \in J : rx = 0\}$, the Ulm invariant at an ordinal α is

$$u_\alpha(K) = \dim_{\mathbf{F}_r} \frac{(r^\alpha K)[r]}{(r^{\alpha+1} K)[r]}. \quad (10)$$

The final classification uses Kiehlmann's countably based dual-reduced theorem only after countability, reducedness, and the complete sequence in (10) have been established [5, Definition 1.3, Theorems 1.4 and 1.8].

3 The exponent embedding and ambient primary factors

Proposition 3.1 (Closed exponent embedding). *The map e_p in (3) is a continuous injection and hence a homeomorphism from $\widehat{\mathbf{Z}}$ onto the closed subgroup H_p .*

Proof. For $\ell \neq p$, the map $\mathbf{Z} \rightarrow \mathbf{Z}_\ell^\times$, $n \mapsto p^n$, is continuous for the profinite topology: the inverse image of an open subgroup contains $d\mathbf{Z}$, where d is the order of p in the associated finite quotient. It therefore extends uniquely to $\widehat{\mathbf{Z}}$. Taking the product proves continuity.

Write $\widehat{\mathbf{Z}} = \prod_r \mathbf{Z}_r$ and examine the pro- r factor. If $r \neq p$ is odd, use the decomposition

$$p = \omega_r(p) \langle p \rangle_r, \quad \omega_r(p) \in \mu_{r-1}, \quad \langle p \rangle_r \in 1 + r\mathbf{Z}_r.$$

The pro- r exponent kills the prime-to- r Teichmueller factor and acts on principal units through multiplication by the nonzero element $\log \langle p \rangle_r$. The logarithm $1 + r\mathbf{Z}_r \rightarrow r\mathbf{Z}_r$ is an isomorphism in the required domain [2, Example 8.15]. Since $\mathbf{Z}_r$ is an integral domain, the kernel is zero. If $r = 2 \neq p$, choose $\epsilon_p \in \{\pm 1\}$ with $u_p = \epsilon_p p \in 1 + 4\mathbf{Z}_2$ and apply the logarithm on $1 + 4\mathbf{Z}_2$; the principal unit is again nontrivial [1, Theorem 2.6 and Remark 2.8].

The missing coordinate $r = p$ needs a different detector. For each $m \geq 1$, apply the Bang–Zsigmondy theorem to $p^{p^m} - 1$. Its three classical exceptions do not occur: the exponent is not 1; when it is 2, one has $p = 2$, $m = 1$, and $p + 1 = 3$, not a power of 2; and 6 is not a prime power. A primitive divisor $\ell \neq p$ therefore satisfies

$$\text{ord}_\ell(p) = p^m. \quad (11)$$

This is exactly the original theorem’s primitive-divisor use, sharpened to its printed theorem page [10, p. 283]. If $p^a = 1$ in every away coordinate, then (11) forces the $\mathbf{Z}_p$ component of a to vanish modulo every p^m . All primary components of a are thus zero. Compactness of $\hat{\mathbf{Z}}$ and Hausdorffness of U_p make the image closed. $\square$

Proposition 3.2 (Ambient primary factors). *For every prime r ,*

$$U_{p,(r)} \cong_{\text{top}} \begin{cases} P_r, & r = p, \\ \mathbf{Z}_r \times P_r, & r \neq p. \end{cases} \quad (12)$$

Consequently,

$$B_p \cong_{\text{top}} \prod_r B_{p,(r)}, \quad B_{p,(r)} = U_{p,(r)} / e_p(\mathbf{Z}_r), \quad (13)$$

and all displayed compact groups are countably based.

Proof. For an odd prime $\ell \neq r$, the pro- r Sylow subgroup of $\mathbf{Z}_\ell^\times$ is cyclic of order $r^{v_r(\ell-1)}$. For each $n \geq 1$, Dirichlet’s theorem applied to the reduced residue class $1 + r^n \pmod{r^{n+1}}$ supplies infinitely many primes ℓ with

$$v_r(\ell - 1) = n \quad (14)$$

[9, Theorem 18.1]. After the finitely many forbidden coordinates are removed, each finite cyclic order still occurs countably infinitely often. The local coordinate $\ell = r$, when present, contributes $\mathbf{Z}_r$ for odd r and

$$\mathbf{Z}_2^\times \cong C_2 \times \mathbf{Z}_2 \quad (15)$$

for $r = 2$. In the latter case the finite C_2 is abstractly absorbed into the already countable family of order-2 factors; its restriction-map contribution will be retained in section 4. The local coordinate is absent precisely when $r = p$, proving (12). Apply lemma 2.2 to the closed subgroup H_p and its quotient to obtain (13). Countability of the coordinate set gives a countable base. $\square$

4 Restriction maps and the local Wieferich defect

Put

$$K_{p,r} = B_{p,(r)}^\vee. \quad (16)$$

Applying lemma 2.1 to (13) gives

$$0 \rightarrow K_{p,r} \rightarrow U_{p,(r)}^\vee \xrightarrow{\text{res}_{p,r}} C_{r^\infty} \rightarrow 0. \quad (17)$$

The arrows matter: surjectivity is extension of a continuous character from the closed subgroup $e_p(\mathbf{Z}_r)$, not a claim that every bounded-order character extends with the same order.

When $r = p$, (17) has the form

$$0 \rightarrow K_{p,p} \rightarrow S_p \xrightarrow{g_p} C_{p^\infty} \rightarrow 0. \quad (18)$$

For $r \neq p$ it becomes

$$0 \rightarrow K_{p,r} \rightarrow S_r \oplus C_{r^\infty} \xrightarrow{\Phi_{p,r}} C_{r^\infty} \rightarrow 0. \quad (19)$$

Definition 4.1 (Wieferich–Ulm defect). For primes p, r , define

$$\kappa_r(p) = \begin{cases} 0, & r = p, \\ v_r(p^{r-1} - 1) - 1, & r \neq p \text{ and } r \text{ is odd}, \\ v_2(p^2 - 1) - 3, & r = 2 \text{ and } p \text{ is odd}. \end{cases} \quad (20)$$

Proposition 4.2 (Normalized off-local map). For $r \neq p$, automorphisms of the local source and target normalize (19) to

$$\Phi(s, z) = g(s) + r^{\kappa_r(p)} z, \quad S_r \oplus C_{r^\infty} \rightarrow C_{r^\infty}. \quad (21)$$

Proof. Suppose first that r is odd. Since $\langle p \rangle_r^{r-1} = p^{r-1}$ and $r-1$ is an r -adic unit, the exact logarithmic valuation is

$$v_r(\log \langle p \rangle_r) = v_r(p^{r-1} - 1).$$

Normalize $1 + r\mathbf{Z}_r$ by a logarithmic generator of valuation 1. The primal map on $\mathbf{Z}_r$ is multiplication by an element of valuation $\kappa_r(p)$, so its Pontryagin dual on C_{r^∞} is multiplication by $r^{\kappa_r(p)}$ up to a unit.

For $r = 2$ and p odd, normalize the principal units by $\log 5$. With $u_p = \epsilon_p p \in 1 + 4\mathbf{Z}_2$,

$$v_2\left(\frac{\log u_p}{\log 5}\right) = v_2(u_p - 1) - 2 = v_2(p^2 - 1) - 3. \quad (22)$$

If $p \equiv 3 \pmod{4}$, the nontrivial sign character in the literal local dual $C_2 \oplus C_{2^\infty}$ restricts to the unique order-2 target element; if $p \equiv 1 \pmod{4}$, it restricts trivially. That contribution is placed in g before the abstract C_2 is absorbed into S_2 . Only the principal Pruefer factor is normalized as multiplication by $2^{\kappa_2(p)}$. $\square$

5 The exceptional primary branch

The case $r = p$ is algebraic and does not use off-local exact-order saturation. The distinction is necessary even when $p = 2$.

Lemma 5.1 (Homogeneous triangular split). *Let*

$$S = \bigoplus_{n \geq 1} S_n, \quad S_n = (C_{p^n})^{(\mathbb{N}_0)}, \quad g : S \rightarrow C_{p^\infty}$$

be a homomorphism. Then

$$S = T \oplus R, \quad T \cong S, \quad R \cong \bigoplus_{n \geq 1} C_{p^n}, \quad g(T) = 0. \quad (23)$$

Moreover,

$$\ker g = T \oplus \ker(g|_R) \cong S. \quad (24)$$

Proof. Fix a basis $(e_{n,j})_{j \geq 0}$ of each S_n . Choose $e_{n,0}$ with image of maximal order. Subgroups of $C_{p^\infty}[p^n]$ are linearly ordered, so $g(e_{n,0})$ generates $g(S_n)$. For $j > 0$, choose $c_{n,j} \in \mathbf{Z}/p^n\mathbf{Z}$ and set

$$g(e_{n,j}) = c_{n,j}g(e_{n,0}), \quad e'_{n,j} = e_{n,j} - c_{n,j}e_{n,0}.$$

This is a triangular change of basis on the algebraic direct sum. The $e'_{n,j}$ have exact order p^n and lie in the kernel. Thus $S_n = T_n \oplus R_n$ with $T_n \cong (C_{p^n})^{(\mathbb{N}_0)}$, $R_n \cong C_{p^n}$, and $g(T_n) = 0$. Taking direct sums gives (23) and the literal equality in (24). The subgroup $\ker(g|_R)$ of a direct sum of cyclic primary groups is itself a direct sum of cyclic groups by Kulikov's theorem [4, Corollary 2, pp. 66–67]. It is countable, so each of its cyclic multiplicities is at most $\mathbb{N}_0$ and is absorbed by the already $\mathbb{N}_0$ copies of every order in T . $\square$

Theorem 5.2 (Diagonal Ulm ledger). *For every prime p ,*

$$K_{p,p} \cong S_p, \quad p^\omega K_{p,p} = 0, \quad (25)$$

and

$$u_n(K_{p,p}) = \aleph_0 \quad (n < \omega), \quad u_\alpha(K_{p,p}) = 0 \quad (\alpha \geq \omega). \quad (26)$$

Consequently,

$$B_{p,(p)} \cong_{\text{top}} P_p, \quad \overline{\text{Tor}(B_{p,(p)})} = B_{p,(p)}, \quad \kappa_p(p) = 0. \quad (27)$$

Proof. Apply lemma 5.1 to the epimorphism g_p in (18). Every element of S_p has finite support, hence no nonzero element lies in $p^m S_p$ for every m . This proves the infinite-height assertion. The countably many copies of $C_{p^{n+1}}$ give the finite invariants. Pontryagin duality yields (27); finite-support torsion points are dense in P_p . $\square$

Remark 5.3 (Withdrawn bounded-character route). Surjectivity of (17) does not imply $g_p(S_p[p^m]) = C_{p^\infty}[p^m]$. For example, $\text{ord}_{17}(2) = 8$ while $v_2(17 - 1) = 4$, so a faithful character of the order-8 subgroup of C_{16} may need an extension of order 16. Furthermore, if $\ell \equiv 1 \pmod{8}$, then quadratic reciprocity makes 2 a square modulo ℓ ; one cannot require $v_2(\text{ord}_\ell(2)) = v_2(\ell - 1) \geq 3$. Thus $p = r = 2$ belongs only to theorem 5.2.

6 Exact-order saturation off the diagonal

Fix $r \neq p$ and let $g : S_r \rightarrow C_{r^\infty}$ be the away-coordinate and sign part of (21).

Lemma 6.1 (One exact coordinate). *If a prime $\ell \neq p, r$ satisfies*

$$v_r(\ell - 1) = m, \quad v_r(\text{ord}_\ell(p)) = m, \quad (28)$$

then

$$g(S_r[r^m]) = C_{r^\infty}[r^m]. \quad (29)$$

Proof. The pro- r Sylow of $\mathbf{F}_\ell^\times$ is C_{r^m} , and the exponent map from $\mathbf{Z}_r$ onto that coordinate is surjective because the r -primary part of p has exact order r^m . Dual restriction therefore embeds the coordinate dual onto $C_{r^\infty}[r^m]$. The reverse inclusion is automatic for a homomorphism on r^m -torsion. $\square$

Lemma 6.2 (Exact-order primes). *For every $r \neq p$ and every $m \geq 1$, there are infinitely many primes $\ell \neq p, r$ satisfying (28).*

Proof. Assume first that r is odd. Set

$$F = \mathbf{Q}(\zeta_r), \quad L = F(p^{1/r}), \quad C_m = \mathbf{Q}(\zeta_{r^{m+1}}).$$

The polynomial $X^r - p$ is Eisenstein at every prime of F above p , so L/F is a nontrivial cyclic Kummer extension of degree r ramified above p . The cyclotomic extension C_m/F is ramified only above r . Consequently,

$$L \cap C_m = F. \quad (30)$$

Choose an automorphism of C_m/F sending $\zeta_{r^{m+1}}$ to $\zeta_{r^{m+1}}^{1+r^m}$ and a nontrivial Kummer automorphism sending $p^{1/r}$ to $\zeta_r p^{1/r}$. Equation (30) makes them compatible in the compositum $E = LC_m$, defining an element $\sigma \in \text{Gal}(E/F)$. The field L is the splitting field of $X^r - p$ over $\mathbf{Q}$, so $E/\mathbf{Q}$ is finite Galois. Apply the qualitative Chebotarev theorem to the conjugacy class of σ in $\text{Gal}(E/\mathbf{Q})$. Every conjugate still fixes F ; its restriction to C_m is unchanged because $\text{Gal}(C_m/\mathbf{Q})$ is abelian, and its restriction to L is a nonidentity element of $\text{Gal}(L/F)$. Hence there are infinitely many unramified rational primes with

this Frobenius class [8, Theorem 28.9]; the stable historical source is Lagarias and Odlyzko [6]. Such a prime splits completely in F , so the relevant residue field is $\mathbf{F}_\ell$. Its cyclotomic component yields $v_r(\ell - 1) = m$, while the nontrivial Kummer component says that p is not an r th power in $\mathbf{F}_\ell^\times$. In a cyclic group whose order has r -valuation m , that is equivalent to $v_r(\text{ord}_\ell(p)) = m$.

For $r = 2$, the prime p is odd. Put $L = \mathbf{Q}(\sqrt{p})$ and $C_m = \mathbf{Q}(\zeta_{2^{m+1}})$. The first field is ramified at the odd prime p and the second only at 2, so $L \cap C_m = \mathbf{Q}$. Combine the nontrivial quadratic automorphism with the cyclotomic exponent $1 + 2^m$. The compositum $LC_m/\mathbf{Q}$ is finite abelian, so Chebotarev applies directly to this combined element and gives infinitely many primes with

$$v_2(\ell - 1) = m, \quad \left(\frac{p}{\ell}\right) = -1.$$

Nonsquareness in the cyclic group $\mathbf{F}_\ell^\times$ is equivalent to $v_2(\text{ord}_\ell(p)) = m$. The argument includes $m = 1$ and is disjoint from the diagonal $p = r = 2$ case. $\square$

Theorem 6.3 (Bounded restriction saturation). *For every $r \neq p$ and every $m \geq 1$,*

$$g(S_r[r^m]) = C_{r^\infty}[r^m]. \quad (31)$$

Proof. Combine lemmas 6.1 and 6.2. The theorem is unconditional and qualitative; it uses neither GRH nor an effective density estimate. Moree's results on divisibility of multiplicative orders form a close comparator but do not supply the simultaneous equalities in (28) [7]. $\square$

7 Kernel-internal roots and the complete Ulm sequence

For $r \neq p$, write $\kappa = \kappa_r(p)$ and

$$K = K_{p,r} = \ker \Phi, \quad \Phi(s, z) = g(s) + r^\kappa z. \quad (32)$$

Applying the triangular construction to $g : S_r \rightarrow C_{r^\infty}$ yields

$$S_r = T \oplus R, \quad T \cong S_r, \quad g(T) = 0, \quad (33)$$

and hence an internal split

$$K = T \oplus K^0, \quad K^0 = \ker(\Phi|_{R \oplus C_{r^\infty}}). \quad (34)$$

This homogeneous summand supplies the finite multiplicities; the next theorem computes infinite height.

Theorem 7.1 (Infinite-height subgroup). *Inside $S_r \oplus C_{r^\infty}$,*

$$r^\omega K_{p,r} = 0 \oplus C_{r^{\kappa_r(p)}}. \quad (35)$$

Proof. If (s, z) lies in $r^m K$ for every m , then s belongs to $\bigcap_m r^m S_r = 0$. The kernel equation then gives $r^\kappa z = 0$, which proves the upper inclusion in (35).

For the reverse inclusion, take $z \in C_{r^\infty}[r^\kappa]$ and fix $m \geq 1$. Choose an ambient Pruefer root w_m with $r^m w_m = z$. Then $r^\kappa w_m \in C_{r^\infty}[r^m]$. By theorem 6.3, choose $s_m \in S_r[r^m]$ satisfying

$$g(s_m) = -r^\kappa w_m. \quad (36)$$

Thus (s_m, w_m) lies in K and $r^m(s_m, w_m) = (0, z)$. The root is now inside the kernel. Since this works for every m , $(0, z) \in r^\omega K$. $\square$

Proposition 7.2 (Complete Ulm ledger). *For every $r \neq p$,*

$$u_n(K_{p,r}) = \aleph_0 \quad (n < \omega). \quad (37)$$

If $\kappa_r(p) > 0$, then

$$u_{\omega+\kappa_r(p)-1}(K_{p,r}) = 1 \quad (38)$$

and every other invariant at an ordinal $\alpha \geq \omega$ is zero. If $\kappa_r(p) = 0$, every invariant at $\alpha \geq \omega$ is zero. In all cases $K_{p,r}$ is countable and reduced.

Proof. The summand $T \cong S_r$ in (34) contains countably many copies of $C_{r^{n+1}}$ for every n , giving the lower bound in (37). The entire ambient group is countable, which gives equality. If $\kappa > 0$, [theorem 7.1](#) yields

$$r^{\omega+j}K = r^j C_{r^\kappa} \cong C_{r^{\kappa-j}} \quad (0 \leq j \leq \kappa), \quad r^{\omega+\kappa}K = 0.$$

The order- r subgroups of consecutive terms agree until the last nonzero step, where the quotient is C_r . This gives (38); the case $\kappa = 0$ is immediate. Finally, a divisible subgroup of K must lie in $r^\omega K$, which is finite. A finite divisible group is zero, so K is reduced. $\square$

8 Torsion closure and primary compact structure

Lemma 8.1 (Torsion-closure annihilator). *Let K be a discrete r -group and $B = K^\vee$. Then*

$$\text{Ann}(\overline{\text{Tor}(B)}) = r^\omega K. \quad (39)$$

Proof. If $x \in r^\omega K$ and $\chi \in B$ has order dividing r^m , write $x = r^m y$; then $\chi(x) = 0$. Thus $r^\omega K$ annihilates torsion and its closure. Conversely, if $x \notin r^\omega K$, choose m with $x \notin r^m K$. A character of the bounded quotient $K/r^m K$ separates the nonzero class of x , and it pulls back to a finite-order point of B that does not annihilate x . This is the direct form of Kiehlmann's torsion-closure/infinite-height correspondence [5, Theorem 1.1]. $\square$

Proof of [theorem 1.1](#). The diagonal case is [theorem 5.2](#), so suppose $r \neq p$. Put $N = r^\omega K_{p,r}$. By [lemma 8.1](#) and Pontryagin duality,

$$A_{p,r}^\vee \cong K_{p,r}/N, \quad (B_{p,(r)}/A_{p,r})^\vee \cong N. \quad (40)$$

Projection onto the S_r coordinate gives an exact sequence

$$0 \rightarrow N \rightarrow K_{p,r} \xrightarrow{\pi_S} S_r \rightarrow 0. \quad (41)$$

Indeed, multiplication by r^κ on C_{r^∞} is surjective, so for every $s \in S_r$ there is z with $r^\kappa z = -g(s)$; the kernel of π_S is precisely $0 \oplus C_{r^\kappa} = N$. Hence $K_{p,r}/N \cong S_r$. Dualizing and using [theorem 7.1](#) in (40) yields

$$A_{p,r} \cong_{\text{top}} P_r, \quad B_{p,(r)}/A_{p,r} \cong_{\text{top}} C_{r^\kappa}.$$

$\square$

Because $B_{p,(r)}$ is characteristic and the closure of its torsion is characteristic, [theorem 1.1](#) gives the intrinsic formula

$$\kappa_r(p) = \log_r [B_{p,(r)} : \overline{\text{Tor}(B_{p,(r)})}]. \quad (42)$$

No coordinate label of U_p appears in (42).

9 Global classification and arithmetic separation

Proof of theorem 1.2. Suppose $B_p \cong_{\text{top}} B_q$. A topological isomorphism carries the unique characteristic pro- r factor onto the pro- r factor and carries the closure of torsion onto the closure of torsion. The indices in (42) therefore agree for every r , proving necessity.

Conversely, suppose $\kappa_r(p) = \kappa_r(q)$ for all r . The discrete duals $K_{p,r}$ and $K_{q,r}$ are countable and reduced. Their finite Ulm invariants are all $\mathbb{N}_0$, and their complete transfinite invariants agree by theorem 5.2 and proposition 7.2. Kiehlmann identifies these full Ulm sequences with the compact torsion sequences and classifies countably based dual-reduced pro- r groups by that complete data [5, Remark 1.6 and Theorem 1.8]. Hence $K_{p,r} \cong K_{q,r}$, and duality gives a homeomorphism $\varphi_r : B_{p,(r)} \rightarrow B_{q,(r)}$. The unrestricted product

$$\prod_r \varphi_r : \prod_r B_{p,(r)} \rightarrow \prod_r B_{q,(r)}$$

is a homeomorphism with inverse $\prod_r \varphi_r^{-1}$. Apply (13) to finish the proof. $\square$

Proof of corollary 1.3. At $r = 11$,

$$2^{10} - 1 = 1023 = 3 \cdot 11 \cdot 31, \quad \kappa_{11}(2) = 0,$$

whereas

$$3^{10} - 1 = 59048 = 8 \cdot 11^2 \cdot 61, \quad \kappa_{11}(3) = 1.$$

Thus

$$B_{2,(11)} / \overline{\text{Tor}(B_{2,(11)})} = 1, \quad B_{3,(11)} / \overline{\text{Tor}(B_{3,(11)})} \cong C_{11}.$$

These characteristic quotients cannot be carried to each other by an isomorphism. $\square$

The example is intrinsic, but it does not prove that the full signature recovers every prime. The classification theorem identifies isomorphism classes with equality classes of $p \mapsto \kappa(p)$. Injectivity of that map is not resolved by the present argument.

10 Marked data, prior subtraction, and theorem scope

The quotient theorem is deliberately unmarked. In the presentation

$$H_p \rightarrow U_p \rightarrow B_p,$$

the dual kernels inherit finite-coordinate supports and conductor bounds. These data are natural for the marked sequence but are not characteristic on the bare group. In every homogeneous block, the triangular construction produces a summand $(C_{r^n})^{(\mathbb{N}_0)}$ in the kernel. Basis vectors coming from different away coordinates can be exchanged by an automorphism while the complement is fixed. Such an automorphism moves the marked support filtration. Pontryagin duality transports it to a bare compact-group automorphism.

The ambient group U_p does retain a simpler marker:

$$U_{p,(r)}^\vee \text{ has a divisible } C_{r^\infty} \iff r \neq p. \quad (43)$$

Equation (43) detects the omitted coordinate of U_p , not an invariant of the quotient B_p . Quotienting by H_p replaces that marker with the complete κ -ledger.

The proof-owned delta lies in the joined chain rather than in any classical ingredient. The diagonal restriction kernel needs a different mechanism from the off-local one. Off the diagonal, exact Frobenius conditions produce bounded saturation, bounded saturation produces roots internal to the kernel, and those roots place the arithmetic depth at the ordinal $\omega + \kappa_r(p) - 1$. Only then is the compact classification theorem in the domain of the standard Ulm result.

Table 1: Source-owned input, proof-owned bridge, and retained ceiling.

Input	Used for	Not supplied by the source
Deninger [3]	compact quotient provenance	packet topology, Sylow/Ulm calculation, or classification
Kiehlmann [5]	torsion closure and final dual-reduced classification	arithmetic kernel, heights, or reduction to one defect
Hill/Kulikov [4]	cyclic decomposition of the diagonal residual subgroup	multiplicities or packet identification
Zsigmondy [10]	omitted-coordinate detection	bounded-character saturation
Dirichlet and Chebotarev [6, 8, 9]	exact congruence classes and Frobenius occurrence	the two intersections, selected automorphisms, or kernel roots
Conrad [1, 2]	logarithm domains, valuations, and the 2-sign split	restriction saturation or Ulm conclusions
Moree [7]	nearest multiplicative-order-divisibility comparator	simultaneous exact double valuation or compact classification

10.1 Owner firewall

Table 2: Statements licensed and excluded by owner.

Owner	Licensed statement	Excluded transfer
Marked $H_p \rightarrow U_p \rightarrow B_p$	coordinate support and exponent restriction map	treating labels or conductor bounds as bare invariants
Bare compact group B_p	theorems 1.1 and 1.2 and corollary 1.3	importing the missing-coordinate marker
Actual packet quotient	no theorem in this paper	compact topology, separation, or B_2/B_3 transfer
Measured enhancement	none	Haar descent or canonical packet measure
Flow/time owner	none	a clock, orbit, or return theorem
Trace/operator/determinant	none	spectrum, explicit formula, determinant, or Route-B credit

10.2 Internal Route-A and Route-B placement

This is a nonnormative governance check against the local project protocols `skills/route-a-evaluator.md` and `skills/route-b-evaluator.md`, both version 0.2.0 in the accompanying snapshot. The labels below are not external scientific claims and award no Route credit.

The repository’s Route-A evaluator asks for a frozen dynamical candidate, arithmetic origin, phase space, dynamics, clock, normalization, determinant convention, and data split before it assigns coordinates A0–A4. The object defined in (3) has arithmetic provenance but no dynamics, clock, orbit ledger, transfer operator, or determinant. No Route-A coordinate is therefore asserted.

Route B is a proof route that requires the arithmetic, periodic-orbit, operator, trace, and determinant sides to belong to the same mathematical object. Its minimum input includes an explicit Hilbert space, inner product, dense domain, boundary conditions, operator action, clock, and normalization. None appears in this paper. Accordingly, Route-B advancement is not authorized, the construction is ROUTE_B_NOT_TESTABLE, and no Hilbert–Polya claim is allowed.

11 Limitations and questions unresolved here

The classification has three exact limits.

First, [theorem 1.2](#) proves

$$B_p \cong_{\text{top}} B_q \iff \kappa(p) = \kappa(q),$$

but it does not prove $p = q$. Universal prime recovery requires injectivity of the arithmetic signature, which is not established in this paper.

Second, the theorem is a statement about bare compact groups. Marked support is destroyed by bare automorphisms, and the ambient omitted-coordinate marker does not survive in the same form. An actual packet quotient, a measured enhancement, or a standardized flow requires a separate owner and separate hypotheses.

Third, the proof uses qualitative Chebotarev. It establishes infinitely many exact-order witnesses at every required depth, but it gives no effective least witness, density estimate, or uniform computational complexity bound. No GRH-dependent strengthening is claimed.

Questions left by the present argument are to determine whether $p \mapsto \kappa(p)$ is injective, describe finite collisions if they exist, and quantify the least exact-order witnesses in [lemma 6.2](#). A separate research program may ask whether a packet construction admits an intrinsic dynamics, trace, or operator. Such a program begins outside the theorem proved here and must pass the Route-A and Route-B gates on its own objects.

12 Conclusion

The compact packet bases B_p admit a complete owner-safe classification. Their pro- r factors all have the same large torsion closure P_r ; the finite characteristic quotient records exactly one local depth $\kappa_r(p)$. On the dual side, that number is the position of the unique possible nonzero transfinite Ulm invariant. Equality of every local depth is both necessary and sufficient for a topological isomorphism of the unrestricted primary products. The calculation separates B_2 and B_3 while explaining why the missing ambient coordinate and marked conductor supports are not legitimate shortcuts.

The result is classification, not universal recovery. It also supplies no flow, trace, operator, or determinant. Keeping those ceilings explicit makes the positive theorem sharper: a standard compact-group invariant, computed through exact arithmetic restriction maps, classifies the entire unmarked family.

A Nonnormative declarative conformance appendix

The accompanying workspace contains an MG11 B00/B01 chain that encodes framing rules, typed foundation records, a bounded integrity relation, and an incidence projection vocabulary. It is included as a nonnormative machine-readable conformance record. It is not used in any proof above.

The incidence schema maps each declared attachment carrier to a four-entry row

$$(\text{attachment id}, \text{rule id}, \text{source id}, \text{target id})$$

in strict root order, without folding or deduplication. Persisted, instance-specific receipt records encode an empty success result; a one-row result (10, 9, 2, 3); an E10 source-kind rejection with no projection; and a two-row result

$$(10, 9, 2, 3), \quad (12, 11, 2, 3)$$

with the concrete order $10 < 12$. The artifacts also bind exact byte slices, hashes, and source-scoped dependency endpoints.

These statements are intentionally limited. The MG11 seed declares no authority, execution, or publication. The terminal receipts keep program execution, runtime, output, publication, public view, global scope, and authority at zero. The files therefore record a declarative schema and persisted, instance-specific receipt records, not an executed checker, a formal proof of the schema’s universal correctness, or evidence for [theorems 1.1](#) and [1.2](#). Three-or-more-row behavior, duplicate-key rejection, and the remaining concrete reject variants are not claimed as fixture coverage.

B Regression firewalls

The proof depends on several negative checks that are easy to erase in a compressed presentation:

1. The dual arrow is $0 \rightarrow B_{p,(r)}^\vee \rightarrow U_{p,(r)}^\vee \rightarrow C_{r^\infty} \rightarrow 0$; the last map is restriction and its surjectivity is character extension.
2. Primitive divisors detect the omitted exponent coordinate but do not give bounded-order character saturation. The diagonal branch is handled by [lemma 5.1](#) and [theorem 5.2](#).
3. The C_2 sign restriction is recorded before the abstract finite factor is absorbed into S_2 .
4. The two valuations in [\(28\)](#) are both required. Divisibility of $\text{ord}_\ell(p)$ alone is insufficient.
5. Roots used in [theorem 7.1](#) are corrected by [\(36\)](#), so they lie inside $K_{p,r}$.
6. The compact quotient uses $\overline{\text{Tor}(B)}$, not the potentially nonclosed raw torsion subgroup.
7. Countability and reducedness are proved before the Kiehlmann classification theorem is invoked.
8. The final homeomorphism is the unrestricted product of primary-factor homeomorphisms, not a labelled coordinate identification.

Data and artifact availability

No new empirical data were generated. For this internal draft, the symbolic proof ledger, internal review records, declarative MG11 artifacts, and LaTeX sources are stored only in the local project snapshot. They have not yet been deposited in a public repository, and no stable archival identifier is asserted.

Ethics declaration

The work is mathematical and documentary. It involves no human participants, animals, personal data, or sensitive datasets; institutional ethics approval is not applicable.

Author contributions

Liang Wang: conceptualization, investigation, formal analysis, methodology, project administration, validation, visualization, and writing. The named author retains responsibility for checking every theorem, citation, and manuscript statement before submission.

Conflict of interest

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AI-assisted preparation disclosure

OpenAI Codex was used on 22 August 2026 to organize author-provided local proof ledgers, assist with source-boundary checks, and prepare the LaTeX draft. The tool was not treated as a mathematical authority, did not supply experimental results, and did not authorize any claim. All mathematical and bibliographic content requires final verification by the author before submission.

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